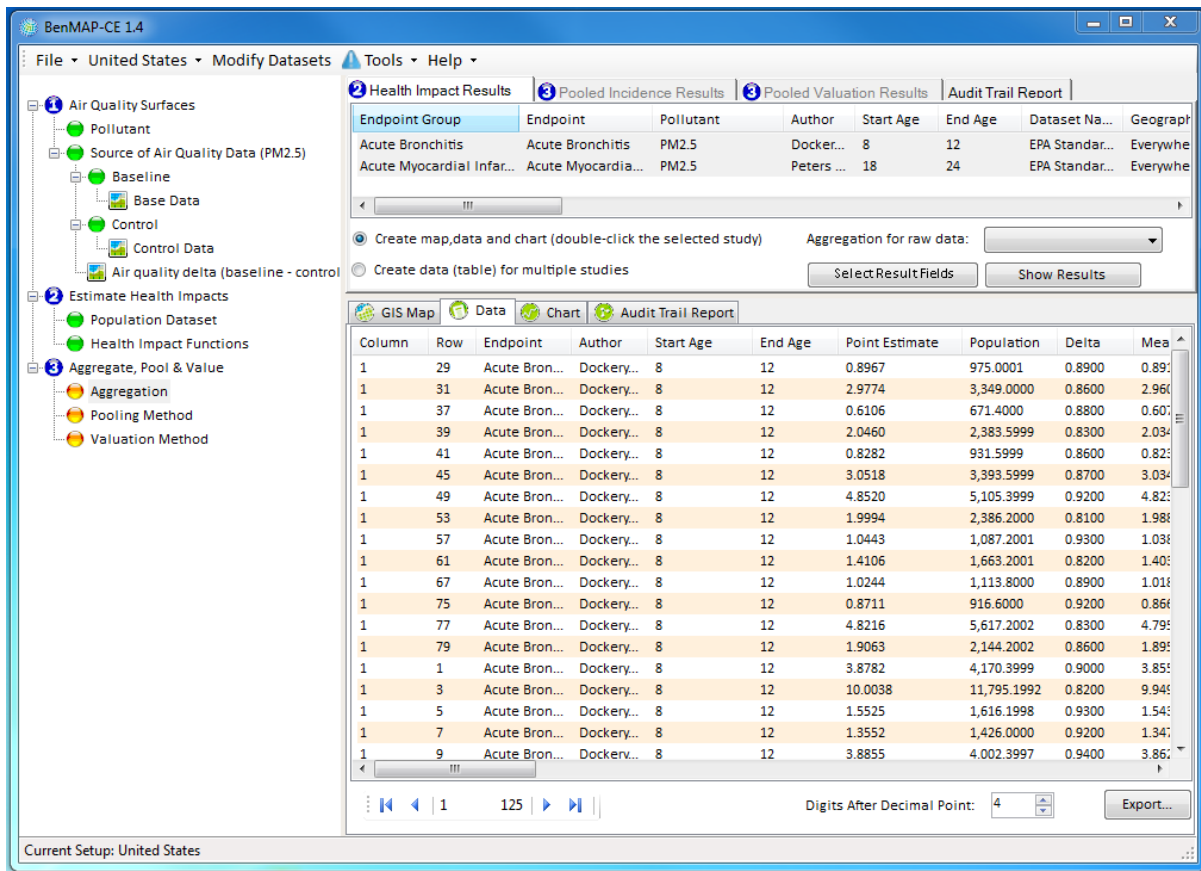




The Health Impact Results tab provides a simple tool to aggregate the raw incidence results. You may select an aggregation level from the **Aggregation for raw data** drop-down list. If you do so, the **Data** table will refresh to display the incidence results aggregated to the selected level.

If you check the option to *Create data (table) for multiple studies*, results will only be available in the **Data** tab. For example, you might want to select three different studies and view aggregated results at the national level. As you modify your choices, the display will be updated accordingly.

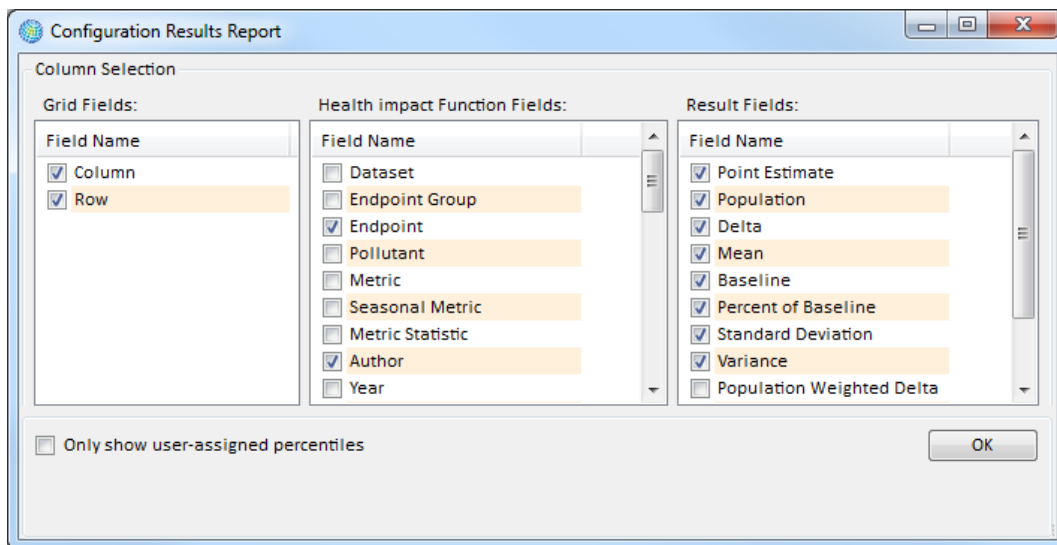
On the **Data** tab, clicking the **Select Result Fields** button opens a **Configuration Results Report** window that allows you to choose the columns that will appear in the results table.

BenMAP-CE Guidance and Best Practices

Below is a list of common culprits that may result in BenMAP-CE calculating a point estimate of zero. If you are generating zero results, begin by double checking the following:

- Did you choose the correct pollutant?
- Does the pollutant definition contain analogous metrics as the health impact function?
- Does the air quality delta (baseline – control) show a change in air quality concentrations? Does it show the necessary pollutant metrics based on the HIF?
- Does the Control scenario have lower air quality values than the Baseline scenario?
- Did you do less than 100% monitor rollback?
- Does the Population Dataset match the location of the Grid Definition?
- Does the Baseline Incidence dataset match the Population Dataset?

- **Grid Fields** permit the inclusion of *Column* and *Row* fields, which can be helpful in identifying the grid-cell of a particular line in the report. For example, when results have been temporarily aggregated to the national level.
- **Health Impact Function Fields** permit the inclusion of various fields which describe or define a function (e.g., *Endpoint Group*, *Endpoint*, *Pollutant*, *Metric*, *Author*, *Year*, *Start Age*, *End Age*, *Gender*, *Race*, *Ethnicity*, *Beta*, *Beta Distribution*). These fields can be helpful in identifying the health impact function associated with a particular line in the report.
- **Result Fields** permit the inclusion of fields associated with results of this analysis (e.g., *Point Estimate*, *Population*, *Delta*, *Mean*, *Percentiles*).



At the bottom of the **Data** tab, there is also an option to specify the number of digits that appear after the decimal point (click the up or down arrows to edit the number beside the **Digits After Decimal Point** field, or type a number directly in the box). Also at the bottom of the **Data** tab is an option to display **All Percentiles** rather than the default 2.5th and 97.5th percentiles.

Table 6-1 provides a summary of the optional fields that have not been previously described in this section.

Table 6-1. Selected Variables in the Reports Based on the APVR file

Variable	Description
Column	The column of the grid cell of the result. For grid cell level results, this is the column of the grid cell. For county and state level results, this is the state FIPS code. For national results, this is always 1.
Row	The row of the grid cell of the result. For grid cell level results, this is the row of the grid cell. For U.S. county results, this is the county FIPS code. For U.S. state and national results, this is always 1.
Dataset	Specifies the dataset from which a health impact function was chosen.
Population	Population provides the number of persons used in the health impact function calculation.
Delta	The difference between the baseline and control scenarios for the metric used in the health impact function. Calculated by subtracting the metric value in the control scenario from the metric value in the baseline scenario.
Point Estimate	The point estimate for the result from the health impact function. The point estimate is generally based on the mean estimate of the "Beta" from the health impact function.
Mean	Mean of the points in the Monte Carlo-generated distribution for this result. The mean is set to missing if the Point Mode option is chosen.
Baseline	Estimate based on the baseline function, which typically estimates health impacts due to all causes (not just air pollution-related causes).
Percent of Baseline	Estimates the percentage change in health impacts (e.g., hospital admissions) due to the change in air quality from the baseline to the control scenario. Calculated by dividing the Point Estimate by the Baseline.
Standard Deviation	Standard deviation calculated based on the points in the Monte Carlo-generated percentiles for this result.
Variance	Variance calculated based on the points in the Monte Carlo-generated percentiles for this result.
Percentiles	The number of percentiles depends on the number of points in Monte Carlo-generated percentiles for this result.

Once all of the options have been selected for your report, you can export the **Health Impact Results** data. First select the **Data** tab then click the **Export** button. This will bring up a window allowing you to name the file you want to save. Note that by default BenMAP-CE will export the file to the CFGR folder. Carefully name the file that you are generating so that you will recognize it in the future!

Decision Point

Once you generate a CFGRX file, BenMAP-CE has quantified the endpoint-specific health benefits associated with your Control scenario. You have completed your BenMAP-CE analysis if:

3. You do not want to pool your health incidence results. For many of the health endpoints (e.g., mortality), BenMAP-CE contains several different health impact functions from different studies that you may choose to include in your configuration. Pooling refers to combining the results of two or more health impact functions into a single, integrated result.
4. You do not want to monetize the health benefits in the valuation step.

You may export your health impact results by selecting your desired aggregation level from the *Aggregation for raw data* drop-down list and clicking the **Export** button from the **Data** tab. If you also want to pool your health incidence results and/or monetize the health benefits, proceed to Ch. 7 "Aggregating, Pooling, and Valuing."

6.5 Frequently Asked Questions

How do I know which health impact functions to use? Which functions does EPA use?

One option regarding the choice of health impact functions is to work with someone, say another BenMAP-CE user, who is familiar with the epidemiological literature and develop your own set of health impact functions. Reviewing the epidemiological literature can be time consuming, though in some situations, this might be the best option. For example, it would be worthwhile to develop health impact functions to estimate the impacts of carbon monoxide exposure, for which BenMAP-CE does not have pre-installed functions.

Another option is to use the ozone and PM_{2.5} configurations used by EPA. These are available on the BenMAP-CE website (<http://www.epa.gov/benmap/benmap-community-edition>). These functions are derived from the epidemiology literature described in the appendices to this user manual. If desired you can edit this configuration and then save it under a different file name—it is always a good idea to keep the original version, so you can go back to it if needed!

How do I edit or add other health impact functions?

To edit or add health impact functions you need to go to **Modify Datasets** window available from the BenMAP-CE main menu. See the health impact function section in Chapter 4: Loading Data for details on how to do this.

How do I learn more about the population data in BenMAP-CE?

Appendix J describes the population data for the *United States* setup in detail.

Why did I not get results for a given geographic area that I wanted in my analysis?

Check to see if your air quality grids mapped properly.